

20 Common Network Ports You Must Know (and Secure)

Printable cheat sheet • inventory, restrict exposure, enforce encryption, and monitor

Tip: a port number is just a rendezvous point — security depends on what's listening, how it's configured, and who can reach it.

Port	Proto	Typical service	Why defenders care	Secure baseline
20	TCP	FTP data	Legacy file transfer, tricky firewalling	Avoid if possible; prefer SFTP/HTTPS; restrict to internal segments
21	TCP	FTP control	Cleartext creds possible; brute force magnet	Disable; if required: FTPS, strong auth, IP allowlist, monitoring
22	TCP	SSH	Administrative access; brute force; key theft	Prefer keys over passwords; MFA where possible; restrict by IP/VPN; harden ciphers
23	TCP	Telnet	Cleartext remote login	Disable everywhere you can; replace with SSH
25	TCP	SMTP relay	Spam abuse; open relay risk	Lock relays; enforce SPF/DKIM/DMARC; rate limit; monitor abuse
53	TCP/UDP	DNS	Redirection; data exfil; amplification if exposed	Restrict recursion; limit zone transfers; log queries; authoritative should answer TCP+UDP
67	UDP	DHCP server	Rogue DHCP = instant network chaos	Enable DHCP snooping; trusted ports only; segment guest networks
68	UDP	DHCP client	Client side of DHCP (often overlooked)	Same as above; watch for unexpected DHCP offers
80	TCP	HTTP	Cleartext; often 'temporary' then forgotten	Redirect to 443; keep 80 only for redirects if needed
110	TCP	POP3	Cleartext mailbox access; legacy auth risk	Avoid; use POP3S (995) or modern auth; disable basic auth where possible
123	UDP	NTP	Time poisoning; reflection if exposed	Use trusted upstreams; restrict queries; prefer authenticated NTP where supported
135	TCP	MS RPC Endpoint Mapper	Gateway to many Windows services; lateral movement	Internal only; restrict by subnet; monitor for scanning/lateral movement
139	TCP	NetBIOS Session Service	Legacy Windows file sharing surface	Disable where possible; block at boundaries; migrate to modern SMB

20 Common Network Ports You Must Know (and Secure)

Printable cheat sheet • inventory, restrict exposure, enforce encryption, and monitor

Tip: a port number is just a rendezvous point — security depends on what's listening, how it's configured, and who can reach it.

Port	Proto	Typical service	Why defenders care	Secure baseline
143	TCP	IMAP	Mailbox access; legacy auth exposure	Prefer IMAPS (993); modern auth; disable basic auth
161	UDP	SNMP	Default community strings; device intel leak	Use SNMPv3; restrict to management VLAN; block from user networks
389	TCP/UDP	LDAP	Cleartext binds; directory exposure	Require TLS/STARTTLS and signing; restrict clients; monitor binds
443	TCP	HTTPS	High-value target; hides in normal traffic	Patch fast; strong TLS; restrict admin paths; monitor auth anomalies
445	TCP	SMB	Ransomware's favorite hallway; wormable misconfigs	Never internet-exposed; restrict internal; disable SMBv1; monitor shares
587	TCP	SMTP submission	Client outbound mail; abuse if weak auth	Require auth + STARTTLS; rate limit; monitor for spikes
3389	TCP	RDP	Credential theft; brute force; ransomware staging	Behind VPN/RD Gateway/ZTNA; MFA; NLA; lockouts; allowlists

Secure Baseline (quick wins)

- Default deny: allow only required inbound ports per system role; block everything else at the edge and between VLANs.
- Keep management off the internet (SSH/RDP/RPC/SMB/SNMP/LDAP): use VPN/RD Gateway/ZTNA + MFA + allowlists.
- Prefer encryption: HTTPS over HTTP, SFTP/FTPS over FTP, IMAPS/POP3S over legacy mailbox ports.
- Bind services to the right interfaces (localhost/private), and disable unused listeners.
- Log and alert: auth failures, new listeners, unusual spikes, and east-west scanning are your early-warning system.

Sources: <https://www.iana.org/assignments/service-names-port-numbers> • <https://csrc.nist.gov/pubs/sp/800/41/r1/final>

© Kyle Reddoch (CybersecKyle) • <https://www.kylereddoch.me>